

- 106.** If $2 * 3 = \sqrt{13}$ and $3 * 4 = 5$, then the value of $5 * 12$ is
 (a) $\sqrt{17}$ (b) $\sqrt{29}$
 (c) 12 (d) 13
- 107.** If 1537* is a perfect square, then the digit which replaces * is (Hotel Management, 2007)
 (a) 2 (b) 4
 (c) 5 (d) 6
- 108.** The smallest perfect square that is divisible by 7! is (I.I.F.T., 2010)
 (a) 19600 (b) 44100
 (c) 176400 (d) 705600
- 109.** The least perfect square number divisible by 3, 4, 5, 6 and 8 is
 (a) 900 (b) 1200
 (c) 2500 (d) 3600
- 110.** The least perfect square, which is divisible by each of 21, 36 and 66, is
 (a) 213444 (b) 214344
 (c) 214434 (d) 231444
- 111.** The least number by which 294 must be multiplied to make it a perfect square, is
 (a) 2 (b) 3
 (c) 6 (d) 24
- 112.** Find the smallest number by which 5808 should be multiplied so that the product becomes a perfect square.
 (a) 2 (b) 3
 (c) 7 (d) 11
- 113.** The least number by which 1470 must be divided to get a number which is a perfect square, is
 (a) 5 (b) 6
 (c) 15 (d) 30
- 114.** What is the smallest number to be subtracted from 549162 in order to make it a perfect square?
 (a) 28 (b) 36
 (c) 62 (d) 81
- 115.** What is the least number which should be subtracted from 0.000326 to make it a perfect square?
 (a) 0.000002 (b) 0.000004
 (c) 0.02 (d) 0.04
- 116.** What is the least number to be added to 7700 to make it a perfect square?
 (a) 77 (b) 98
 (c) 131 (d) 221
 (e) None of these (Bank Recruitment, 2008)
- 117.** The smallest number to be added to 680621 to make the sum a perfect square is (S.S.C., 2005)
 (a) 4 (b) 5
 (c) 6 (d) 8
- 118.** The greatest four-digit perfect square number is
 (a) 9000 (b) 9801
 (c) 9900 (d) 9981
- 119.** The least number of 4 digits which is a perfect square is
 (a) 1000 (b) 1016
 (c) 1024 (d) 1036
- 120.** The sum of 18 consecutive natural numbers is a perfect square. What is the smallest possible value of this sum? (A.A.O. Exam, 2009)
 (a) 169 (b) 225
 (c) 289 (d) 441
- 121.** $\sqrt{2+\sqrt{3}} \cdot \sqrt{2+\sqrt{2+\sqrt{3}}} \cdot \sqrt{2+\sqrt{2+\sqrt{2+\sqrt{3}}}}$
 $\cdot \sqrt{2-\sqrt{2+\sqrt{2+\sqrt{3}}}}$ is equal to
 (a) 1 (b) 2
 (c) 4 (d) $\sqrt{6}$
- 122.** Given $\sqrt{5} = 2.2361$, $\sqrt{3} = 1.7321$, then $\frac{1}{\sqrt{5}-\sqrt{3}}$ is equal to
 (a) 1.98 (b) 1.984
 (c) 1.9841 (d) 2
- 123.** $\frac{1}{(\sqrt{9}-\sqrt{8})} - \frac{1}{(\sqrt{8}-\sqrt{7})} + \frac{1}{(\sqrt{7}-\sqrt{6})} - \frac{1}{(\sqrt{6}-\sqrt{5})}$
 $+ \frac{1}{(\sqrt{5}-\sqrt{4})}$ is equal to (M.B.A., 2007)
 (a) 0 (b) $\frac{1}{3}$
 (c) 1 (d) 5
- 124.** Determine the value of
 $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \dots + \frac{1}{\sqrt{120}+\sqrt{121}}$.
 (a) 8 (b) 10
 (c) $\sqrt{120}$ (d) $12\sqrt{2}$ (M.A.T., 2005)
- 125.** The expression $1 - \frac{1}{1+\sqrt{3}} + \frac{1}{1-\sqrt{3}}$ equals
 (a) $1-\sqrt{3}$ (b) 1
 (c) $-\sqrt{3}$ (d) $\sqrt{3}$ (M.B.A., 2011)
- 126.** $\left(2 + \sqrt{2} + \frac{1}{2+\sqrt{2}} + \frac{1}{\sqrt{2}-2}\right)$ simplifies to
 (a) $2-\sqrt{2}$ (b) 2
 (c) $2+\sqrt{2}$ (d) $2\sqrt{2}$ (M.B.A., 2007)